

Modeling with First-order DEs

3.1 Linear Models

In this section, we will present some applications of 1st order DEs

ex: ① Suppose that the number of bacteria in a culture grows at a rate proportional to the number of bacteria present at time t . After 3 hours, there are 400 bacteria and after 10 hours, there are 2000 bacteria. How many bacteria was there initially?

Let $P = P(t)$ = # of bacteria at time t measured in hours.

$$\frac{dP}{dt}$$

↙
The rate of change of P

$$\propto$$

↓
is proportional to

$$P$$

↓
the number of bacteria

$$\frac{dP}{dt} \propto P$$

$$\Rightarrow \frac{dP}{dt} = kP \quad \text{for a constant } k$$

Let's solve this DE. It is separable.

$$\Rightarrow \int \frac{dP}{P} = \int k \, dt$$

$$\Rightarrow \ln P = kt + C$$

Note: $P > 0$ so we don't need an absolute-value.

$$\Rightarrow P = e^{kt+C} = e^C e^{kt}$$

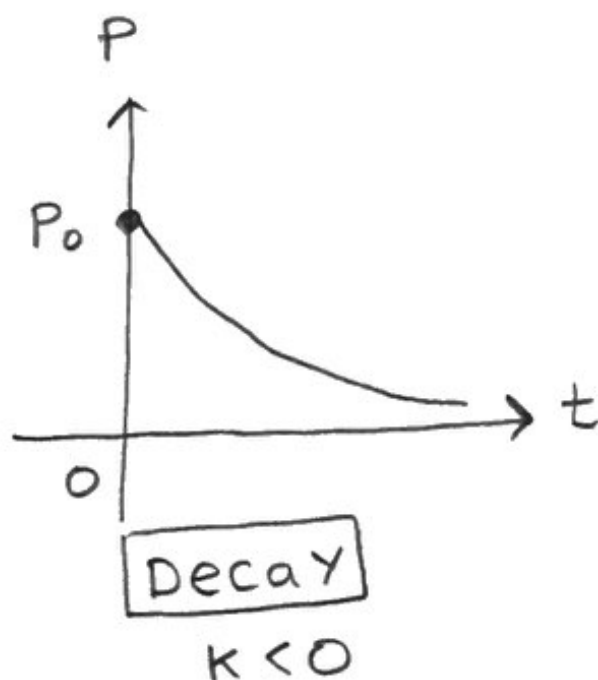
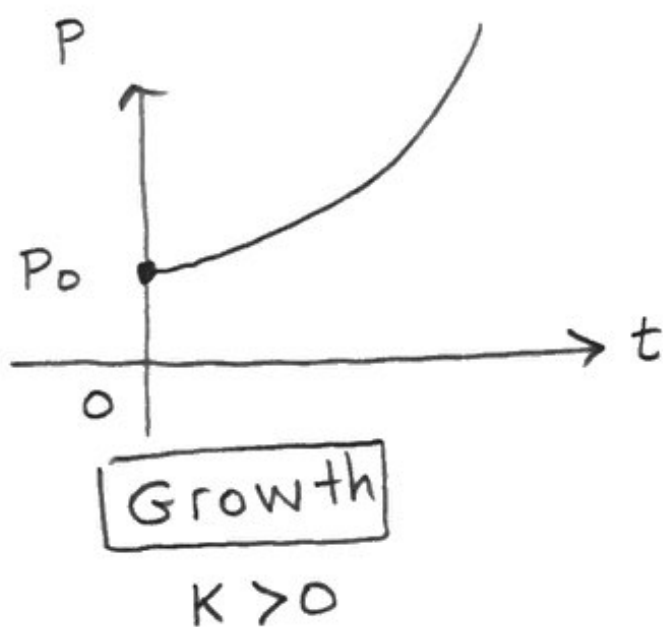
Suppose that at $t=0$, $P = P_0$

P_0 = initial number of bacteria.

$$\Rightarrow P_0 = e^C e^0 \Rightarrow e^C = P_0$$

$$\Rightarrow \boxed{P = P_0 e^{kt}}$$

$P = P_0 e^{kt}$ is an equation that models exponential growth (if $k > 0$) or exponential decay (if $k < 0$)



Our bacteria example is an example of exponential growth.

The amount of a Radiactive substance is an example of exponential decay.

Back to our bacteria problem.

@ $t = 3$ hours $\rightarrow P = 400$ bacteria

@ $t = 10$ hours $\rightarrow P = 2000$ bacteria

$$\Rightarrow \begin{cases} P_0 e^{3k} = 400 \\ P_0 e^{10k} = 2000 \end{cases}$$

$$\Rightarrow \frac{P_0 e^{10k}}{P_0 e^{3k}} = \frac{2000}{400}$$

$$\Rightarrow e^{7k} = 5$$

$$\Rightarrow 7k = \ln 5$$

$$\Rightarrow k = \frac{1}{7} \ln 5 > 0 \quad (\text{Growth})$$

$$P_0 = 400 e^{-3k}$$

$$\Rightarrow P_0 = 400 e^{-3\left(\frac{\ln 5}{7}\right)}$$

$$\Rightarrow \boxed{P_0 \approx 200 \text{ bacteria}}$$

Initially, there was about
200 bacteria

ex: ② Newton's Law of cooling

states that : " The temperature of an object changes at a rate proportional to the difference between its temperature and the environment temperature. "

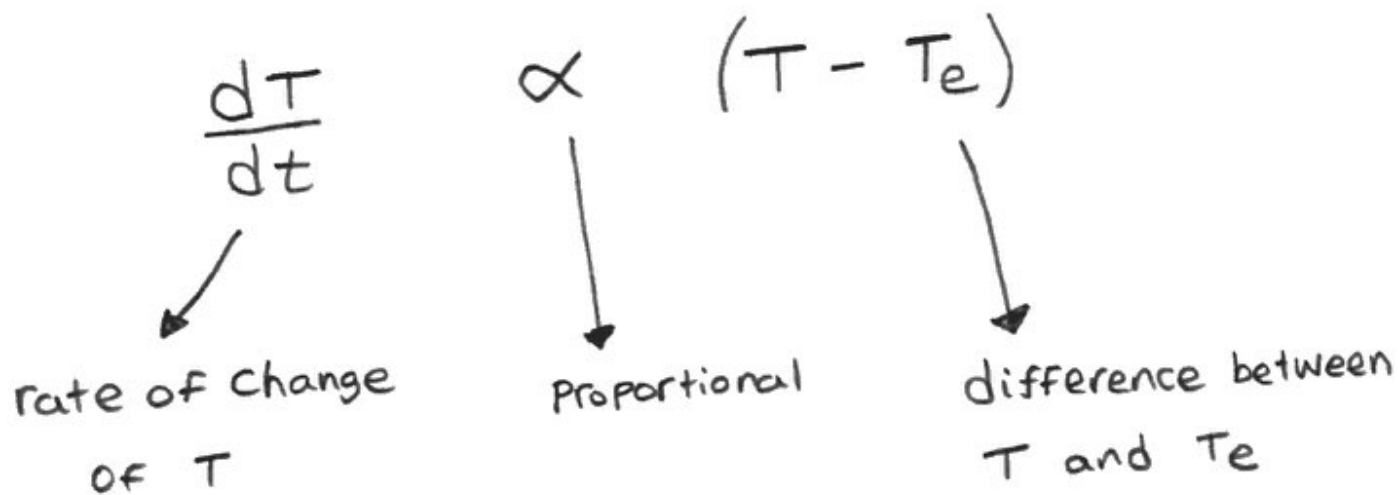
Suppose a pot of soup was initially at a temperature of 100°C . Five minutes later its temperature was 70°C .

How much more time is needed for the soup to have a temperature of 50°C ? Assume the soup is in a room with a constant temperature of 20°C .

Let $T = T(t)$ = temperature of the object (the soup) at time t

T_e = environment temperature

Newton's Law of cooling



Note : We are translating the words into mathematical concepts.

$$\Rightarrow \frac{dT}{dt} = -K (T - T_e)$$

for a constant $K > 0$.

$$\Rightarrow \frac{dT}{dt} + kT = kT_e$$

This is a Linear DE.

$$\text{I.F. } e^{\int k dt} = e^{kt}$$

$$\Rightarrow \underbrace{e^{kt} \frac{dT}{dt} + k e^{kt} T}_{\text{"product rule"}} = k e^{kt} T_e$$

$$\Rightarrow \frac{d}{dt} [e^{kt} T] = k e^{kt} T_e$$

$$\Rightarrow e^{kt} T = \int k e^{kt} T_e dt$$

Note: In the general case, T_e can depend on time. Here, we assume that T_e is constant.

$$\Rightarrow e^{kt} T = k T_e \int e^{kt} dt$$

$$\Rightarrow e^{kt} T = k T_e \left[\frac{e^{kt}}{k} \right] + C$$

$$\Rightarrow e^{kt} T = T_e e^{kt} + C$$

$$\Rightarrow \boxed{T = T_e + C e^{-kt}}$$

Note : observe that $\lim_{t \rightarrow \infty} T = T_e$.

In the long run, the object (the soup) will be at the room temperature. That makes sense.

Pot of soup

$$100^{\circ}\text{C} \xrightarrow{5 \text{ minutes}} 70^{\circ}\text{C} \xrightarrow{(\text{?})} 50^{\circ}\text{C}$$

Room: $T_e = 20^{\circ}\text{C}$

$$@ t = 0, T = 100$$

$$100 = 20 + c e^0 \rightarrow \boxed{c = 80}$$

$$@ t = 5, T = 70$$

$$70 = 20 + 80 e^{-5k}$$

$$\Rightarrow 50 = 80 e^{-5k}$$

$$\Rightarrow e^{5k} = 8/5$$

$$\Rightarrow \boxed{k = \frac{1}{5} \ln(8/5)}$$

Find t when $T = 50^\circ\text{C}$

$$50 = 20 + 80 e^{-kt}$$

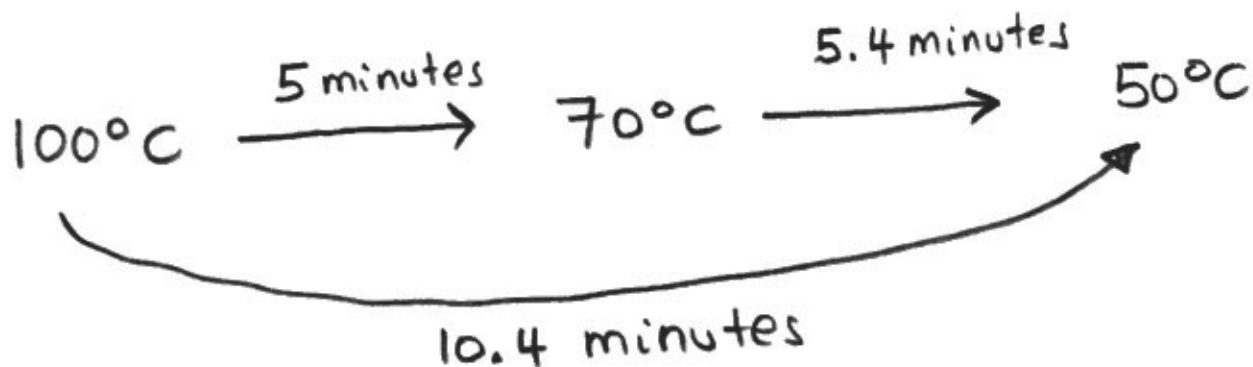
$$30 = 80 e^{-kt}$$

$$e^{kt} = 8/3$$

$$\Rightarrow t = \frac{1}{k} \ln(8/3)$$

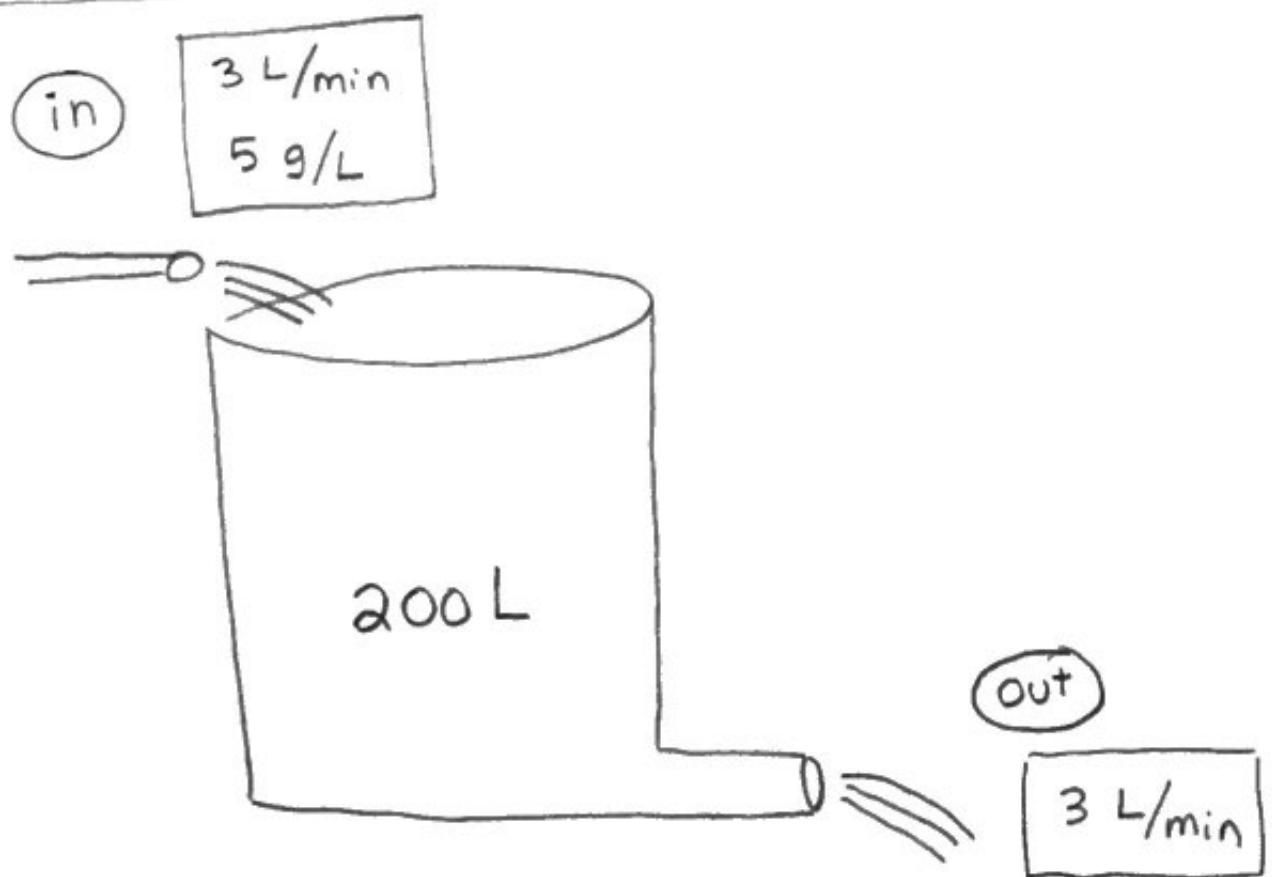
Plug $k = \frac{1}{5} \ln(8/5)$ to get:

$$t = 10.4 \text{ minutes}$$



Answer: Another 5.4 minutes is needed.

ex: (3) A tank contains 200 L of fluid in which 50g of salt is dissolved initially. Brine containing 5 g/L of salt is pumped into the tank at a rate of 3 L/min. The well-mixed solution is pumped out at the same rate. After how much time will there be 100g of salt in the tank?



Let $m = m(t)$ = mass of salt in the tank at time t .

$$\frac{dm}{dt} = (\text{Rate in}) - (\text{Rate out})$$

↓
rate of
change of m

Let's express each rate in grams/minutes.

$$\frac{dm}{dt} = \underbrace{\left(5 \frac{\text{g}}{\text{L}}\right) \left(3 \frac{\text{L}}{\text{min}}\right)}_{\text{Rate in}} - \underbrace{\left(\frac{m}{200} \frac{\text{g}}{\text{L}}\right) \left(3 \frac{\text{L}}{\text{min}}\right)}_{\text{Rate out}}$$

Note: $\frac{m}{200} \text{ g/L}$ is the concentration of the fluid coming out. There is m grams of salt dissolved into 200 L of fluid.

$$\Rightarrow \frac{dm}{dt} = 15 - \frac{3m}{200}$$

$$\Rightarrow \frac{dm}{dt} + \underbrace{\left(\frac{3}{200}\right)}_{P(t)} m = 15 \quad \leftarrow \text{Linear DE}$$

$$\text{I.F. } e^{\int P(t) dt} = e^{\int \frac{3}{200} dt} = e^{3t/200}$$

$$\Rightarrow \frac{d}{dt} \left[m e^{3t/200} \right] = 15 e^{3t/200}$$

$$\Rightarrow m e^{3t/200} = 15 \int e^{3t/200} dt$$

$$\Rightarrow m e^{3t/200} = 15 \left[\frac{e^{3t/200}}{3/200} \right] + C$$

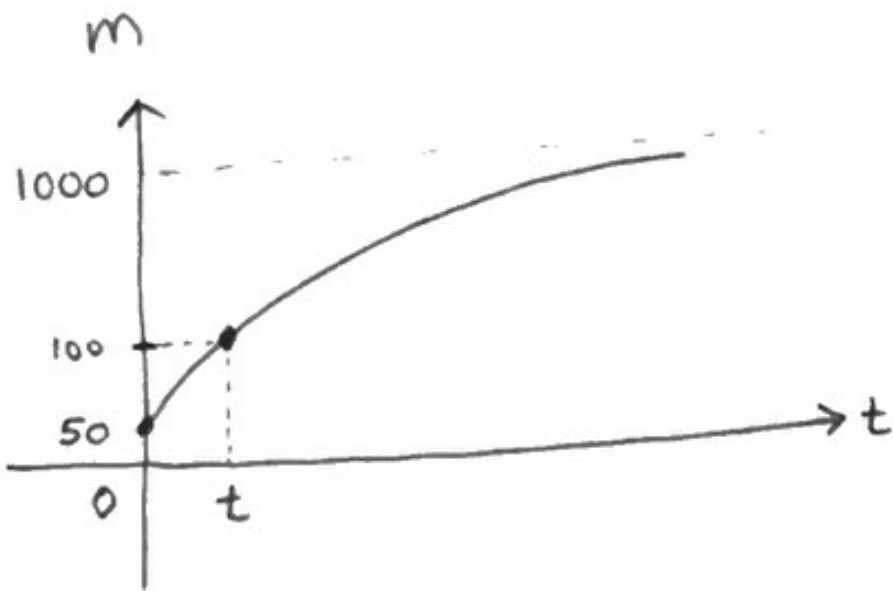
$$\Rightarrow m = 1000 + C e^{-3t/200}$$

Initially there is 50g of salt in the tank.

$$m(0) = 50 \text{ g}$$

$$50 = 1000 + C e^0 \rightarrow C = -950$$

$$\Rightarrow \boxed{m = 1000 - 950 e^{-3t/200}}$$



We want to find t when $m = 100 \text{ g}$.

Note : $\lim_{t \rightarrow \infty} m = 1000 \text{ g}$

$$100 = 1000 - 950 e^{-3t/200}$$

$$\Rightarrow 950 e^{-3t/200} = 900$$

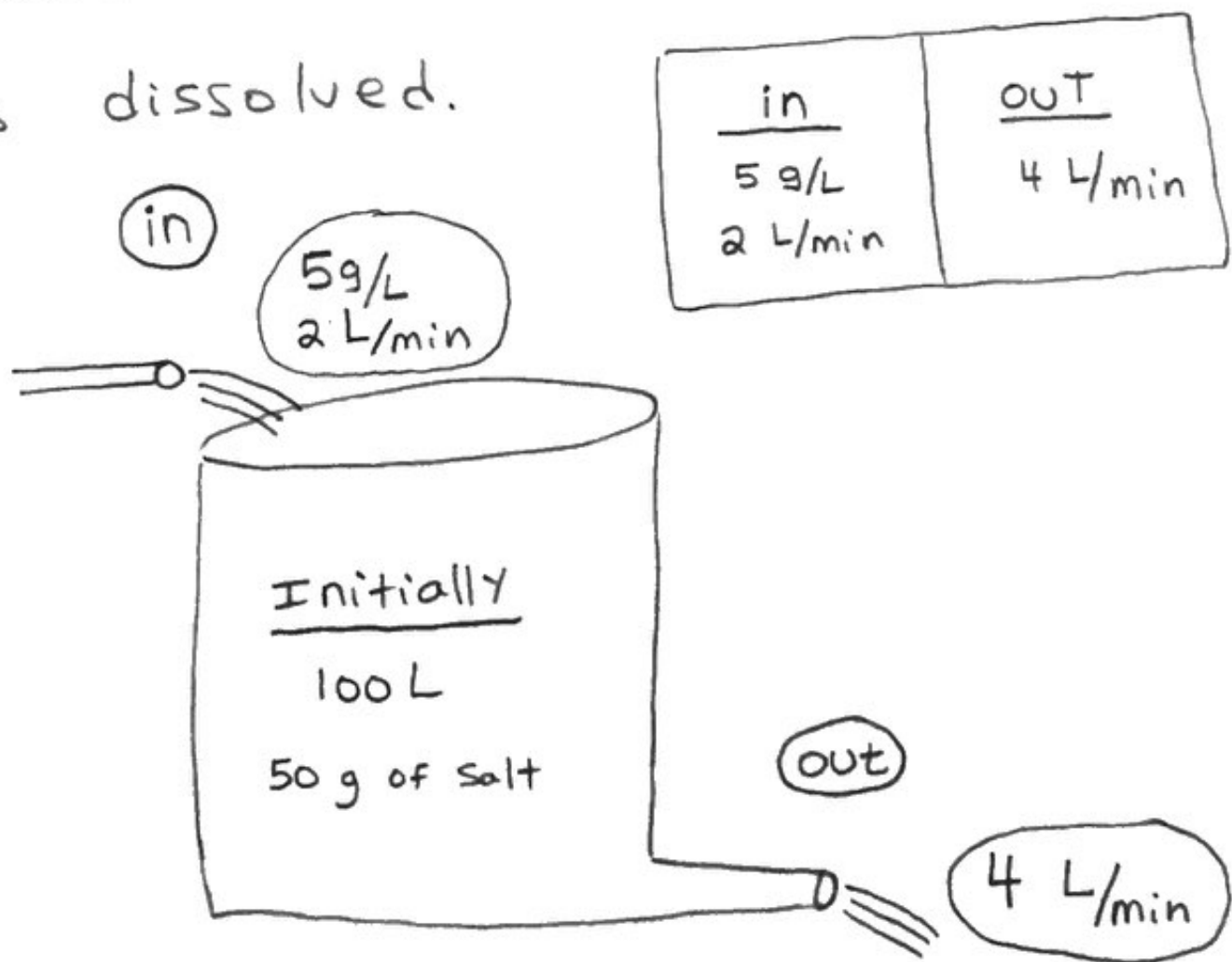
$$\Rightarrow e^{-3t/200} = \frac{900}{950}$$

$$\Rightarrow t = -\frac{200}{3} \ln\left(\frac{90}{95}\right)$$

$$\Rightarrow t \approx 3.6 \text{ minutes}$$

ex: ④ Suppose we modify the previous example so that the volume of fluid in the tank changes with time.

Assume the tank contains initially 100 L of fluid in which 50g of is dissolved.



$V = V(t)$ = Volume of fluid in the tank
at time t .

$$\begin{aligned}\frac{dv}{dt} &= (\text{Rate in}) - (\text{Rate out}) \\ &= 2 \text{ L/min} - 4 \text{ L/min}\end{aligned}$$

$$\frac{dv}{dt} = -2 \Rightarrow V = -2t + C$$

$$@ t = 0, V = 100 \text{ L} \Rightarrow C = 100$$

$$\therefore \boxed{V = 100 - 2t}$$


the volume varies with time.

$m = m(t)$ = mass of salt in the tank
at time t .

$$\frac{dm}{dt} = (\text{Rate in}) - (\text{Rate out})$$

$$\frac{dm}{dt} = \underbrace{\left(5 \frac{\text{g}}{\text{L}}\right) \left(2 \frac{\text{L}}{\text{min}}\right)}_{\text{in}} - \underbrace{\left(\frac{m}{V} \frac{\text{g}}{\text{L}}\right) \left(4 \frac{\text{L}}{\text{min}}\right)}_{\text{out}}$$

$$\frac{dm}{dt} = 10 - \frac{4m}{100-2t}$$

$$\frac{dm}{dt} = 10 - \frac{2}{50-t}$$

$$\Rightarrow \boxed{\frac{dm}{dt} + \frac{2}{50-t} m = 10}$$

↑
Linear DE

$$\text{I.F. } e^{\int \frac{2}{50-t} dt} = e^{-2 \ln(50-t)} \\ = (50-t)^{-2}$$

$$\Rightarrow \frac{d}{dt} \left[m(50-t)^{-2} \right] = 10(50-t)^{-2}$$

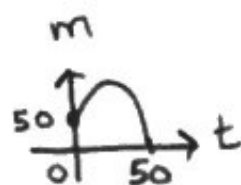
$$\Rightarrow m(50-t)^{-2} = 10(50-t)^{-1} + C$$

$$\Rightarrow m = 10(50-t) + C(50-t)^2$$

$$\text{@ } t=0, m=50 \text{ g}$$

$$\Rightarrow 50 = 10(50) + C(50)^2$$

$$\Rightarrow C = -9/50$$



$$\Rightarrow \boxed{m = 10(50-t) - \frac{9}{50}(50-t)^2}$$

for $0 \leq t \leq 50$. At $t=50$, the tank is empty.

ex: ⑤ The rate of change of a rabbit population P is proportional to the square root of P . The initial population is 100 rabbits. Initially, the rabbits population is increasing at a rate of 20 rabbits per month. How many rabbits will there be one year later?

$$\frac{dP}{dt} \propto \sqrt{P}$$

$$\Rightarrow \frac{dP}{dt} = K \sqrt{P}$$

for a constant K .

@ $t = 0$, $P = 100$ rabbits

and $\frac{dP}{dt} = 20$ rabbits/month

$$\Rightarrow 20 = k \sqrt{100}$$

$$\Rightarrow 20 = 10k$$

$$\Rightarrow k = 2$$

$$\frac{dP}{dt} = 2\sqrt{P} \Rightarrow \int \frac{1}{\sqrt{P}} dP = \int 2 dt$$

$$\Rightarrow \int P^{-1/2} dP = 2t + C_1$$

$$\Rightarrow \frac{P^{1/2}}{1/2} = 2t + C_1$$

$$\Rightarrow 2\sqrt{P} = 2t + C_1$$

$$\Rightarrow \sqrt{P} = t + C$$

Note : $\frac{C_1}{2} = C$

$$\begin{aligned} \text{@ } t=0, P=100 &\Rightarrow \sqrt{100} = 0 + C \\ &\Rightarrow C = 10 \end{aligned}$$

$$\sqrt{P} = t + 10 \Rightarrow P = (t + 10)^2$$

In one year, $t = 12$ months

$$\Rightarrow P = (12 + 10)^2$$

$$\Rightarrow \boxed{P = 484 \text{ rabbits}}$$